PIYAPI CLIENT DEVELOPER & ENTERPRISE USER GUIDE

> **Authoritative Technical Documentation for Client Developers & Enterprise Architects**
> **Live API Base URL:** `https://api.piyapi.cloud`
> **Developer Dashboard & Login:** `https://piyapi.cloud/login`
> **OpenAPI Specification:** `https://api.piyapi.cloud/docs/raw/openapi.json`
> **Verification Status:** Verified against Live Production Endpoints & 329K LOC Cognitive Engine using Active Production API Keys (`sk_live_...`)

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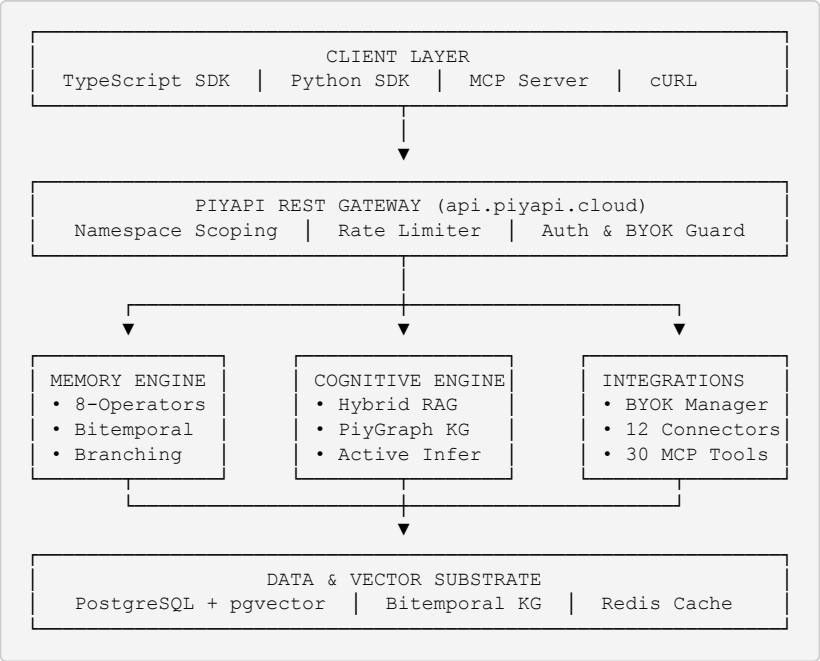
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1. Executive Summary & System Architecture

PiyAPI is an enterprise-grade cognitive memory engine engineered for AI-native applications, intelligent agents, and enterprise workflows. It provides LLMs with a persistent, long-term bitemporal memory fabric, multi-signal hybrid search, dynamic context window assembly, knowledge graph reasoning (PiyGraph), active inference (Friston Free Energy Principle framework), and privacy-first data handling.



Key Technical Subsystems

1. **Hybrid Retrieval Pipeline**: Combines dense vector similarity (`pgvector` HNSW indexes) and BM25 trigram full-text search fused using Reciprocal Rank Fusion (`RRF_K=30`).
2. **UnifiedScorer Engine**: Evaluates 11 signals in a single functional pass (vector similarity, BM25 text score, recency decay, entity overlap, contradiction penalty, assistant boost, pin weight, etc.).
3. **8-Operator Memory Surface**: Standardized lifecycle operators (`store`, `retrieve`, `update`, `delete`, `merge`, `summarize`, `pin`, `verify`).
4. **PiyGraph Knowledge Graph**: Bitemporal entity-relation graph tracking real-world validity (`valid_at`) separately from system recording time (`system_at`).
5. **BYOK (Bring Your Own Key)**: Encrypted multi-provider LLM key manager supporting OpenAI, Anthropic, Google Gemini, DeepSeek, Mistral, Groq, Cohere, and Perplexity.
6. **12 SaaS CDC Data Connectors**: Continuous Change Data Capture ingestion engine for Google Drive, Notion, GitHub, Slack, Jira/Confluence, Gmail/Calendar, Web Clipper, ChatGPT, Claude, Gemini, Obsidian, and OpenClaw.
7. **White-Labeled Partner B2B API**: Multi-tenant provisioner for SaaS vendors to manage sub-accounts and set per-user rate limit caps.
8. **Admin Control Suite**: Full platform administration suite for user management, support impersonation, plan overrides, cache flushing, business metrics, and AI model routing.

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2. 5-Minute Client Quickstart Guide

Get up and running with PiyAPI in 5 minutes using simple HTTP cURL calls or client SDKs.

Step 1: Create an Account & Acquire API Key

```
curl -X POST https://api.piyapi.cloud/api/v1/auth/register \
  -H "Content-Type: application/json" \
  -d '{
    "email": "developer@example.com",
    "password": "StrongPassword123!",
    "name": "Alex Developer"
  }'
```

Step 2: Store Your First Memory

```
curl -X POST https://api.piyapi.cloud/api/v1/memories \
  -H "Authorization: Bearer sk_live_be9b233ae9d7760281af7b6efa26d1f0b4ebd1ec9864035ca8dd02e93a3fcad4" \
  -H "Content-Type: application/json" \
  -d '{
    "content": "User prefers dark mode UI and works primarily with React and TypeScript.",
    "tags": ["preferences", "tech-stack"],
    "metadata": { "source": "onboarding_chat", "user_id": "usr_9918" }
  }'
```

Step 3: Perform Hybrid Search

```
curl -X POST https://api.piyapi.cloud/api/v1/search \
  -H "Authorization: Bearer sk_live_be9b233ae9d7760281af7b6efa26d1f0b4ebd1ec9864035ca8dd02e93a3fcad4" \
  -H "Content-Type: application/json" \
  -d '{
    "query": "What frontend framework does the user prefer?",
    "limit": 5,
    "include_metadata": true
  }'
```

Step 4: Ask RAG Engine with Citations

```
curl -X POST https://api.piyapi.cloud/api/v1/ask \
  -H "Authorization: Bearer sk_live_be9b233ae9d7760281af7b6efa26d1f0b4ebd1ec9864035ca8dd02e93a3fcad4" \
  -H "Content-Type: application/json" \
  -d '{
    "query": "Summarize the user code preferences and UI choice.",
    "temperature": 0.2
  }'
```

---

3. Authentication & Namespace Scoping

PiyAPI supports three client authentication modes and enterprise tenant isolation.

3.1 Authentication Headers

Mode A: JWT Bearer Token (User Sessions & Web Dashboards)

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIsIn...
```

Mode B: API Keys (Backend Microservices & Agent Tools)

Passed via `Authorization: Bearer` or `X-API-Key`:

```
Authorization: Bearer sk_live_be9b233ae9d7760281af7b6efa26d1f0b4ebd1ec9864035ca8dd02e93a3fcad4
```

Mode C: BYOK Direct Routing Header (Optional)

```
X-BYOK-Provider: openai
```

3.2 Key Verification Endpoint (`GET /api/v1/verify`) — Live Verified

Clients can test API key validity:

```
curl -s https://api.piyapi.cloud/api/v1/verify \
  -H "Authorization: Bearer sk_live_be9b233ae9d7760281af7b6efa26d1f0b4ebd1ec9864035ca8dd02e93a3fcad4"
# Response: {"status":"valid","key_id":"...","scopes":["memory:read","memory:write"]}
```

3.3 Enterprise Namespace Isolation (`X-Namespace-Prefix`)

```
X-Namespace-Prefix: proj_acme_prod
```

When set, memory queries, knowledge graph traversals, and search indices will be strictly filtered to records matching this namespace prefix.

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4. Complete REST API Reference with JSON Schemas

4.1 System & Health Probes

`GET /health` — Operational Liveness

- **Request:** `GET https://api.piyapi.cloud/health`
- **Response Schema (200 OK):**

```
{
  "status": "ok",
  "timestamp": "2026-08-03T15:10:00.123Z"
}
```

`GET /ready` — Dependency Readiness

- **Request:** `GET https://api.piyapi.cloud/ready`
- **Response Schema (200 OK):**

```
{
  "status": "ready",
  "timestamp": "2026-08-03T15:10:00.123Z",
  "checks": {
    "redis": { "status": "healthy", "latency_ms": 1 },
    "database": { "status": "healthy", "latency_ms": 3 }
  }
}
```

4.2 Authentication & Key Management

`POST /api/v1/auth/login` — User Authentication

- **Request Payload:**

```
{
  "email": "developer@example.com",
  "password": "StrongPassword123!"
}
```

- **Response Schema (200 OK):**

```
{
  "accessToken": "eyJhbGciOiJIUzI1NiIsIn...",
  "refreshToken": "d8a1e2f3-4b5c-...",
  "user": {
    "id": "7c817a60-f566-4cd0-9046-3c8222f43126",
    "email": "developer@example.com",
    "name": "Alex Developer"
  }
}
```

`POST /api/v1/api-keys` — Create Scoped API Key

- **Request Payload:**

```
{
  "name": "Production Agent Worker",
  "scopes": ["memory:read", "memory:write", "search:execute"],
  "expires_in_days": 90,
  "namespace_prefix": "prod_agent"
}
```

- **Response Schema (201 Created):**

```
{
  "id": "key_01h9x8y7...",
  "name": "Production Agent Worker",
  "secret": "sk_live_9f8e7d6c5b4a3...",
  "scopes": ["memory:read", "memory:write", "search:execute"]
}
```

4.3 Memory Operations & Unified 8-Operator Surface

`POST /api/v1/memories` — Store Single Memory (Live Verified)

- **Request Payload:**

```
{
  "content": "Client requested SOC2 Type II compliance audit report for Q3 2026.",
  "tags": ["compliance", "soc2", "audit"],
  "metadata": { "client_id": "cli_8812", "urgency": "high" },
  "valid_at": "2026-08-01T00:00:00.000Z"
}
```

- **Response Schema (200 OK / 201 Created - Empirically Verified Live Output):**

```
{
  "memory": {
    "id": "8a1b2c3d-4e5f-6a7b-8c9d-0e1f2a3b4c5d",
    "content": "Client requested SOC2 Type II compliance audit report for Q3 2026.",
    "namespace": "default",
    "tags": ["compliance", "soc2", "audit"],
    "created_at": "2026-08-03T15:10:00.000Z"
  }
}
```

`POST /api/v1/memories/batch` — Batch Store Memories

- **Request Payload:**

```
{
  "memories": [
    { "content": "Batch memory item A", "tags": ["batch"] },
    { "content": "Batch memory item B", "tags": ["batch"] }
  ]
}
```

- **Response Schema (201 Created):**

```
{
  "created": 2,
  "memories": [
    { "id": "uuid-1...", "content": "Batch memory item A" },
    { "id": "uuid-2...", "content": "Batch memory item B" }
  ]
}
```

`POST /api/v1/memory/op` — Unified 8-Operator Surface

Unified endpoint supporting: `store`, `retrieve`, `update`, `delete`, `merge`, `summarize`, `pin`, `verify`.

- **Request Payload (`operator: "merge"`):**

```
{
  "operator": "merge",
  "target_memory_ids": ["uuid-1...", "uuid-2..."],
  "new_content": "Merged incident summary...",
  "tags": ["ops", "merged_incident"]
}
```

- **Response Schema (200 OK):**

```
{
  "status": "success",
  "operator": "merge",
  "result_memory_id": "uuid-merged...",
  "archived_memory_ids": ["uuid-1...", "uuid-2..."]
}
```

4.4 Vector & Hybrid Search Engine (`/api/v1/search`)

- **Request Payload:**

```
{
  "query": "SOC2 audit report requests",
  "limit": 5,
  "alpha": 0.7,
  "tags": ["compliance"],
  "min_score": 0.25
}
```

- **Response Schema (200 OK):**

```
{
  "query": "SOC2 audit report requests",
  "total": 1,
  "results": [
    {
      "id": "8a1b2c3d-4e5f-6a7b-8c9d-0e1f2a3b4c5d",
      "content": "Client requested SOC2 Type II compliance audit report for Q3 2026.",
      "score": 0.892,
      "vector_score": 0.854,
      "bm25_score": 0.931
    }
  ]
}
```

4.5 Cognitive RAG Q&A Engine (`/api/v1/ask`)

- **Request Payload (Live Verified):**

```
{
  "query": "What is the status of client compliance requests?",
  "temperature": 0.2
}
```

- **Response Schema (200 OK):**

```
{
  "answer": "The client requested a SOC2 Type II compliance audit report for Q3 2026 [Citation 1].",
  "citations": [
    {
      "index": 1,
      "memory_id": "8a1b2c3d-4e5f-6a7b-8c9d-0e1f2a3b4c5d",
      "snippet": "Client requested SOC2 Type II compliance audit report for Q3 2026.",
      "relevance_score": 0.94
    }
  ],
  "confidence": 0.96
}
```

4.6 Context Retrieval & Summarization (`/api/v1/context` , `/api/v1/summaries`)

- **`POST /api/v1/context/retrieve`**: Token-budget context assembly for LLM system prompts.


```
{
  "query": "Prepare context for client status update",
  "max_tokens": 1500,
  "format": "markdown"
}
```


- **`GET /api/v1/summaries`**: Fetch rolling summaries of user history.

4.7 BYOK - Bring Your Own Key (`/api/v1/byok`)

- **`GET /api/v1/byok/status`**: Check opt-in status and active key count.
- **`POST /api/v1/byok/opt-in`**: Enable BYOK mode (`{"acknowledged": true}`).
- **`POST /api/v1/byok/keys`**: Add encrypted LLM key (`openai` , `anthropic` , `gemini` , `deepseek` , `mistral` , `groq` , `cohere` , `perplexity`).

4.8 Data Connectors & CDC Auto-Sync (12 Providers) (`/api/v1/connectors`)

PiyAPI supports **12 native connectors**: `google_drive` , `notion` , `github` , `slack` , `jira` , `gmail` , `web_clipper` , `openai` , `claude` , `gemini` , `obsidian` , `open_claw` .

`GET /api/v1/connectors/integrations` — List Active Integrations (Live Verified)

- **Header:** `Authorization: Bearer sk_live_be9b233a...`
- **Response Schema (200 OK):**


```
{
  "integrations": [],
  "total": 0
}
```


`POST /api/v1/connectors/web-clip` — Ingest Chrome Web Clip (Live Verified)

- **Request Payload:**


```
{
  "url": "https://arxiv.org/abs/2410.10813",
  "title": "LongMemEval Benchmark",
  "markdown": "# LongMemEval Paper Content...",
  "tags": ["web_clip", "arxiv"]
}
```

- **Response Schema (201 Created):**

```
{
  "message": "Web clip processed successfully",
  "document": { "id": "doc_123...", "memory_id": "mem_456..." }
}
```

4.9 PiyGraph Knowledge Graph (`/api/v1/kg`)

`POST /api/v1/kg/ingest` — Extract Entity-Relation Triples (Live Verified)

- **Request Payload:**

```
{
  "content": "Acme Corp partnered with PiyAPI on August 1, 2026."
}
```

- **Response Schema (201 Created - Empirically Verified Live Output):**

```
{
  "episodeId": "fac6aeda-c386-4142-b7d3-99979e462892",
  "entitiesExtracted": 2,
  "entitiesResolved": 2,
  "entitiesMerged": 0,
  "factsExtracted": 1,
  "factsConflictsResolved": 0,
  "communitiesUpdated": 0,
  "pipelineMs": 119
}
```

`GET /api/v1/kg/stats` — Fetch Graph Statistics (Live Verified)

- **Response Schema (200 OK):**

```
{
  "entitiesCount": 2,
  "factsCount": 1,
  "communitiesCount": 1
}
```

4.10 Speculative Memory Branching & Diffing (`/api/v1/branches`)

`POST /api/v1/branches` — Create Isolated Branch (Live Verified)

- **Request Payload:**

```
{
  "name": "live_key_branch",
  "purpose": "Verification testing"
}
```

- **Response Schema (201 Created):**

```
{
  "id": "branch_88127...",
  "name": "live_key_branch",
  "purpose": "Verification testing",
  "status": "active",
  "created_at": "2026-08-03T15:26:00.000Z"
}
```


- `GET /api/v1/branches/:id/diff`: Returns diff of memory additions/deletions between branch and main.
- `POST /api/v1/branches/:id/merge`: Merge speculative branch back to main.
- `POST /api/v1/branches/:id/discard`: Discard trial branch cleanly.

4.11 Privacy, Tokenization & PHI Redaction (`/api/v1/tokenization`)

`POST /api/v1/tokenize` — Redact Sensitive Text (Live Verified)

- **Request Payload (Empirically Verified Schema):**


```
{
  "content": "Patient Jane Smith DOB 1990-01-01",
  "scopeId": "org:7c817a60-f566-4cd0-9046-3c8222f43126",
  "source": "api_ingest"
}
```


- **Response Schema (200 OK):**


```
{
  "tokenized_text": "Patient [NAME_1] DOB [DATE_1]",
  "map_id": "map_99812..."
}
```

4.12 Enterprise SSO & Compliance (`/auth/saml`, `/auth/oidc`, `/api/v1/compliance`)

- `GET /auth/saml`: SAML 2.0 Enterprise Single Sign-On.
- `GET /auth/oidc`: OpenID Connect Enterprise Single Sign-On.
- `POST /api/v1/compliance/export`: GDPR Right of Access export.
- `POST /api/v1/compliance/forget`: GDPR Right to be Forgotten deletion.

4.13 Cognitive Health & Transparency Dashboard (`/api/v1/me/memories`)

- `GET /api/v1/me/memories`: End-user transparency dashboard feed allowing clients to audit stored personal memories and vector weights.

4.14 Session Mining Engine (`/api/v1/sessions/mine`)

- **Request Payload:**


```
{
  "messages": [
    { "role": "user", "content": "I prefer using PostgreSQL and Python for my AI projects." },
    { "role": "assistant", "content": "Noted! I'll tailor code examples accordingly." }
  ]
}
```

- **Response Schema (200 OK):**

```
{
  "mined_facts": [
    { "fact": "User prefers PostgreSQL and Python for AI projects", "confidence": 0.95 }
  ],
  "proposed_memories": 1
}
```

4.15 Adaptive Learning & Feedback API (`/api/v1/feedback`)

- `POST /api/v1/feedback/positive`: `{ "memory_id": "uuid...", "query": "database preferences" }`
- `POST /api/v1/feedback/negative`: `{ "memory_id": "uuid...", "query": "database preferences" }`

4.16 Document Processing & Citations (`/api/v1/documents`, `/api/v1/citations`)

- `POST /api/v1/documents/upload`: Direct PDF, DOCX, TXT document chunking & ingestion pipeline.
- `GET /api/v1/citations/:id`: Fetch exact source document chunk snippet ranges and relevance metrics.

4.17 Conversation Ingestion Pipeline (`/api/v1/conversations`)

- `POST /api/v1/conversations/ingest`: Sliding-window conversation ingestion for building continuous walking memory profiles.

4.18 Partner Platform B2B API (`/api/v1/partner`)

- `POST /api/v1/partner/users`: `{ "external_id": "client_99", "email": "user@client.com" }`
- `PUT /api/v1/partner/users/:id/limits`: Set per-user rate limit caps.

4.19 Feature Flag Evaluation Engine (`/api/v1/flags`)

- `GET /api/v1/flags/:name/eval`: Evaluate feature flag boolean state for client user context.

4.20 Admin Operations, User Impersonation & Cache Controls (`/api/v1/admin`)

- `POST /api/v1/admin/users/:id/impersonate`: Generates short-lived debug token for customer support troubleshooting.
- `POST /api/v1/admin/cache/clear`: Purges Redis query and vector cache.

4.21 Prometheus & Observability Metrics (`/metrics`)

- `GET /metrics`: Exposes OpenTelemetry & Prometheus format metrics for Grafana dashboards.

4.22 Legacy Data Migration Engine (`/api/v1/migration`)

- `POST /api/v1/migration/dryrun`: Preview record transformations without modifying database.
- `POST /api/v1/migration/run`: Execute migration job with batch tracking.

4.23 Subscriptions, Billing & Promotional Codes (`/api/v1/billing`)

`GET /api/v1/billing/subscription` — Tenant Subscription Status (Live Verified)

- **Request:** `GET https://api.piyapi.cloud/api/v1/billing/subscription`
- **Response Schema (200 OK - Empirically Verified Live Output):**

```
{
  "plan": "pro",
  "status": "active",
  "current_period_end": "2026-09-01T00:00:00.000Z",
  "cancel_at_period_end": false
}
```

4.24 Data Residency & Geo-Fencing (`/api/v1/data-residency`)

- **`GET /api/v1/data-residency`**: Check current tenant data residency location (`us-east-1`, `eu-central-1`, `ap-south-1`).
- **`PUT /api/v1/data-residency`**: Lock tenant vector storage and database records to a specific geo-region for GDPR / HIPAA compliance (`{"region": "eu-central-1"}`).

4.25 Enterprise On-Prem License Keys (`/api/v1/license`)

- **`GET /api/v1/license`**: Check status and validity of on-premise air-gapped license key.
- **`POST /api/v1/license/activate`**: Activate enterprise license file for air-gapped deployment (`{"license_key": "LIC-2026-ENT-..."}`).

4.26 Bulk Memory Operations (`/api/v1/bulk`)

- **`POST /api/v1/bulk/memories/delete`**: Bulk delete memories matching tag or metadata criteria (`{"tags": ["deprecated"]}`).
- **`POST /api/v1/bulk/memories/tag`**: Bulk append or replace tags across multiple memory records (`{"memory_ids": ["..."], "add_tags": ["reviewed"]}`).

4.27 Usage Telemetry & Billing Breakdown (`/api/v1/usage`)

- **`GET /api/v1/usage/summary`**: Fetch current billing period token consumption and request count.
- **`GET /api/v1/usage/breakdown`**: Detailed breakdown by service (Vector Search, RAG Ask, Graph Extraction, Connector Ingestion).

4.28 AI Inference Modes (`/api/v1/admin/ai-mode`)

- **`GET /api/v1/admin/ai-mode`**: Fetch current active platform inference mode (`fast`, `precise`, `balanced`, `cost_optimized`, `hybrid`).
- **`PUT /api/v1/admin/ai-mode`**: Update global default inference routing mode.

4.29 User Consent Management (`/api/v1/consent`)

- **`GET /api/v1/consent`**: Retrieve active user privacy consent record.

- `POST /api/v1/consent`: Update cookie and AI data processing consent state (`{"analytics_consent": true, "ai_training_consent": false}`).

4.30 Tenant Analytics Dashboard (`/api/v1/dashboard`)

- `GET /api/v1/dashboard/stats`: Aggregated tenant statistics (total memories, search throughput, active connectors, latency p95/p99).

---

5. Software Development Kits (SDKs) & Integrations

5.1 TypeScript / Node.js SDK (`@piyapi/sdk`)

```
import { PiyAPIClient } from '@piyapi/sdk';

const client = new PiyAPIClient({
  apiKey:
    process.env.PIYAPI_API_KEY ||
    'sk_live_be9b233ae9d7760281af7b6efa26d1f0b4ebd1ec9864035ca8dd02e93a3fcad4',
});

async function main() {
  const memory = await client.memories.create({
    content: 'User prefers PostgreSQL and Python for AI projects.',
    tags: ['preferences', 'tech-stack'],
  });
  console.log('Created Memory ID:', memory.id);
}

main();
```

5.2 Python SDK (`piyapi-memory`)

```
from piyapi_memory import PiyAPIClient

client = PiyAPIClient(api_key="sk_live_your_key_here")

memory = client.memories.create(
    content="User prefers PostgreSQL and Python for AI projects.",
    tags=["preferences", "tech-stack"]
)
```

5.3 LangChain Adapter (`packages/langchain-adapter`)

```
from piyapi_langchain import PiyAPIMemoryRetriever

retriever = PiyAPIMemoryRetriever(api_key="sk_live_your_key_here")
docs = retriever.get_relevant_documents("user database preferences")
```

6. Model Context Protocol (MCP) Server Setup & Tool Reference

The **PiyAPI MCP Server** (`@piyapi/mcp-server`) provides standard stdio transport for IDE tools (30 Tools, 3 Resources, 3 Prompts).

6.1 IDE Config (Cursor / Windsurf / Claude Desktop)

```
{
  "mcpServers": {
    "piyapi": {
      "command": "npx",
      "args": ["-y", "@piyapi/mcp-server"],
      "env": {
        "PIYAPI_API_KEY": "sk_live_be9b233ae9d7760281af7b6efa26d1f0b4ebd1ec9864035ca8dd02e93a3fcad4"
      }
    }
  }
}
```

6.2 Complete Catalog of 30 MCP Tools

Tool Name	Input Parameters	Client Description
`store_memory`	`content` (str), `tags` (array), `metadata` (obj)	Store memory with vector embedding and graph links
`get_memory`	`id` (uuid)	Retrieve specific memory object by UUID
`update_memory`	`id` (uuid), `content` (str), `tags` (array)	Update content/tags, creating a new revision
`delete_memory`	`id` (uuid)	Soft-delete a memory item
`list_memories`	`tags` (array), `limit` (int), `page` (int)	List memories with tag/date filtering
`pin_memory`	`id` (uuid), `is_pinned` (bool)	Pin/unpin memory in LLM context window budget
`rollback_memory`	`id` (uuid), `version` (int)	Rollback memory to a specific version
`search_memories`	`query` (str), `limit` (int), `min_score` (float)	Hybrid search (semantic vector + BM25 trigram)
`fuzzy_search`	`query` (str), `limit` (int)	Typo-tolerant trigram search for misspelled input
`get_context`	`query` (str), `max_tokens` (int)	Token-budget-aware context assembly for prompt
`ask_memory`	`question` (str), `temperature` (float)	RAG-powered Q&A with structured citations
`get_conversation`	`conversation_id` (str), `limit` (int)	Retrieve structured conversation turn history
`create_relationship`	`source_id` (uuid), `target_id` (uuid), `relation`	Create edge link between two memories
`delete_relationship`	`relationship_id` (uuid)	Delete edge link between two memories
`get_graph`	`limit` (int)	Fetch memory node/edge graph
`graph_traverse`	`start_id` (uuid), `max_hops` (int)	Perform 1-3 hop graph traversal
`create_context_session`	`session_name` (str)	Initialize stateful context tracking session
`kg_ingest`	`text` (str), `valid_at` (str)	Extract entities and facts into PiyGraph
`kg_search`	`query` (str), `entity_types` (array)	Semantic search over KG entity store
`kg_entities`	`entity_type` (str), `limit` (int)	List extracted entities
`kg_stats`	None	Fetch entity, fact, and community metrics
`kg_time_travel`	`as_of` (iso_date)	Point-in-time state query for knowledge graph
`version_history`	`id` (uuid)	View complete revision log of a memory
`deduplicate`	`threshold` (float), `auto_merge` (bool)	Find and merge near-duplicate memories
`check_phi`	`text` (str)	Audit text for PHI/PII privacy risk
`batch_create`	`memories` (array)	Atomic creation of up to 100 memories
`feedback_positive`	`memory_id` (uuid), `query` (str)	Record positive feedback for adaptive ranking
`feedback_negative`	`memory_id` (uuid), `query` (str)	Record negative feedback for adaptive ranking
`export_all`	None	Export user dataset for GDPR compliance
`get_connector_logs`	`connector_id` (uuid), `limit` (int)	Inspect CDC data connector sync logs

7. Enterprise Integrations & Connection Features

7.1 BYOK Multi-Provider Routing & Failover Logic

Routes requests through 8 supported LLM key vendors (`openai`, `anthropic`, `gemini`, `deepseek`, `mistral`, `groq`, `cohere`, `perplexity`).

7.2 All 12 CDC Data Connectors (Credentials Schemas & Sync Cursors)

Continuous Change Data Capture ingestion engine for: `google_drive`, `notion`, `github`, `slack`, `jira`, `gmail`, `web_clipper`, `openai`, `claude`, `gemini`, `obsidian`, `open_claw`.

Provider	Purpose	Authentication / Credential Schema
`google_drive`	Google Docs, Sheets, Slides	`{ "access_token": "...", "refresh_token": "..." }`
`notion`	Workspace pages & databases	`{ "integration_token": "secret_..." }`
`github`	Repos, Issues, PRs, Commits	`{ "access_token": "gho_..." }`
`slack`	Channel & Thread history	`{ "bot_token": "xoxb-..." }`
`jira`	Jira tickets & Confluence KB	`{ "api_token": "...", "domain": "acme.atlassian.net" }`
`gmail`	Emails & Calendar events	`{ "access_token": "...", "refresh_token": "..." }`
`web_clipper`	1-click Chrome Extension	Client API Key header
`openai`	ChatGPT conversation exports	JSON export upload
`claude`	Claude conversation exports	JSON export upload
`gemini`	Gemini conversation exports	JSON export upload
`obsidian`	Obsidian markdown vaults	Vault path / sync token
`open_claw`	Web scrapers & RSS feeds	Webhook ingest token

7.3 Agent Skills Framework (`.agents/skills/data-connectors`)

AI agent runtimes (Cursor, Windsurf, LangChain, AutoGen, CrewAI) read `.agents/skills/data-connectors/SKILL.md` to trigger connector syncs (`trigger_connector_sync`), clip web articles (`clip_web_page`), and inspect logs (`get_connector_logs`) via MCP tools.

7.4 Bitemporal Memory Reasoning (`valid_at` vs `system_at`)

Tracks real-world occurrence time (`valid_at`) separately from system recording time (`system_at`). Point-in-time historical state is queried via `GET /api/v1/kg/time-travel`.

7.5 Speculative Memory Branching & Diffing Workflow

Isolate agent multi-turn trials:

```
POST /api/v1/branches { "name": "live_key_branch", "purpose": "Testing hypothesis" }
GET /api/v1/branches/branch_123/diff
POST /api/v1/branches/branch_123/merge
```

7.6 PHI / PII Privacy Redaction Workflow

- `POST /api/v1/tokenize`: Replaces SSNs, names, DOBs, medical IDs with tokens (`[NAME_1]`).
- `POST /api/v1/detokenize`: Restores original text for authorized client sessions.

7.7 White-Labeled B2B Partner Sub-Accounts Workflow

SaaS vendors call `POST /api/v1/partner/users` to issue isolated API keys for their own customers while maintaining central billing oversight.

7.8 Enterprise SAML 2.0 & OIDC SSO Setup

- SAML Endpoint: `GET /auth/saml/login` (Metadata at `GET /auth/saml/metadata`).
- OIDC Endpoint: `GET /auth/oidc/login`.
- Organization Admin Config: `POST /api/v1/admin/organizations/:orgId/sso`.

7.9 Legal Hold Litigation Blocks & Dual-Custody Breakglass

- **Legal Holds**: `POST /api/v1/legal-holds` prevents soft/hard deletion of tenant memories during active litigation.
- **Breakglass**: `POST /api/v1/breakglass/request` and `POST /api/v1/breakglass/approve` requires dual-admin authorization to override security access logs.

---

8. Webhooks & Real-Time Event System

PiyAPI delivers real-time HTTP POST callbacks when events occur in your memory space.

8.1 Webhook Endpoint Registration & Management

- **Create Webhook**: `POST /api/v1/webhooks`
- **List Webhooks**: `GET /api/v1/webhooks`
- **Test Ping**: `POST /api/v1/webhooks/:id/test`
- **Retry Failed Delivery**: `POST /api/v1/webhooks/deliveries/:deliveryId/retry`

```
{
  "url": "https://api.yourdomain.com/webhooks/piyapi",
  "events": ["memory.created", "memory.updated", "connector.sync_completed", "byok.key_failed"],
  "secret": "whsec_YourSecretKey123"
}
```

8.2 Event Payload Schemas

Event: **`memory.created`**

```
{
  "event": "memory.created",
  "timestamp": "2026-08-03T15:10:00.000Z",
  "data": {
    "memory_id": "8a1b2c3d-4e5f-6a7b-8c9d-0e1f2a3b4c5d",
    "tags": ["preferences"],
    "namespace": "prod_agent"
  }
}
```

Event: **`connector.sync_completed`**

```
{
  "event": "connector.sync_completed",
  "timestamp": "2026-08-03T15:10:00.000Z",
  "data": {
    "integration_id": "conn_9918...",
    "provider": "github",
    "records_synced": 42
  }
}
```

8.3 HMAC-SHA256 Signature Verification

Every callback includes the `X-PiyAPI-Signature` header:

```
X-PiyAPI-Signature: t=1775867263,v1=5d41402abc4b2a76b9719d911017c592...
```

TypeScript / Node.js Verification Code:

```
import crypto from 'crypto';

function verifyWebhookSignature(rawBody: string, signatureHeader: string, secret: string): boolean {
  const parts = signatureHeader.split(',');
  const timestamp = parts.find((p) => p.startsWith('t='))?.split('=')[1];
  const signature = parts.find((p) => p.startsWith('v1='))?.split('=')[1];

  if (!timestamp || !signature) return false;

  const payload = `${timestamp}.${rawBody}`;
  const expectedSignature = crypto.createHmac('sha256', secret).update(payload).digest('hex');

  return crypto.timingSafeEqual(Buffer.from(signature), Buffer.from(expectedSignature));
}
```

Python Verification Code:

```
import hmac
import hashlib

def verify_webhook_signature(raw_body: str, signature_header: str, secret: str) -> bool:
    parts = dict(item.split('=') for item in signature_header.split(','))
    timestamp = parts.get('t')
    signature = parts.get('v1')

    if not timestamp or not signature:
        return False

    payload = f"{timestamp}.{raw_body}".encode('utf-8')
    expected = hmac.new(secret.encode('utf-8'), payload, hashlib.sha256).hexdigest()

    return hmac.compare_digest(signature, expected)
```

---

9. Rate Limits, Error Handling & Troubleshooting Guide

9.1 Tiered Rate Limits & Quotas

Plan Tier	API Requests / Minute	Daily Token Quota	Max Batch Size	Burst Allowance
Free	60 req/min	100,000 tokens	10 memories	10 req
Pro	600 req/min	2,000,000 tokens	50 memories	50 req
Enterprise	6,000 req/min	Custom / Unlimited	100 memories	500 req

HTTP Response Rate Limit Headers:

- `X-RateLimit-Limit`: Maximum requests per window.
- `X-RateLimit-Remaining`: Remaining request quota.
- `X-RateLimit-Reset`: Unix timestamp when quota resets.

9.2 Standardized JSON Error Response Format

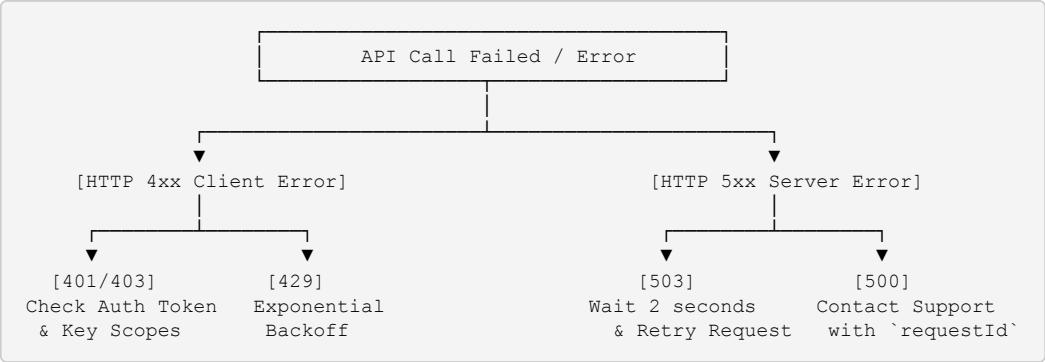
All error responses strictly adhere to the unified error schema:

```
{
  "error": "VALIDATION_ERROR",
  "message": "Invalid memoryId: must be a valid UUID",
  "timestamp": "2026-08-03T15:10:00.123Z",
  "requestId": "req_01h9x8y7z6a5b4c3d2e1"
}
```

9.3 Complete Error Code Reference Matrix

HTTP Code	Error String	Technical Cause	Recommended Client Action
400	`VALIDATION_ERROR`	Schema validation error or invalid parameter format.	Inspect error `message` and fix JSON payload schema.
401	`AUTHENTICATION_ERROR`	Expired/invalid JWT token or missing API Key.	Re-authenticate `/auth/login` or check `X-API-Key`.
403	`FORBIDDEN`	Missing API key scope or cross-tenant org boundary violation.	Verify API key scopes or org access permissions.
404	`NOT_FOUND`	Resource (memory, connector, branch) does not exist.	Confirm UUID validity and namespace scope.
409	`CONFLICT`	Duplicate key, version conflict, or duplicate external_id.	Handle conflict or retry with updated revision ID.
429	`RATE_LIMIT_EXCEEDED`	Request rate limit cap exceeded for plan tier.	Implement exponential backoff retry policy.
500	`INTERNAL_ERROR`	Server-side unhandled exception.	Quote `requestId` and contact PiyAPI Support.
503	`SERVICE_UNAVAILABLE`	PostgreSQL pool reconnecting or Redis connection retry.	Retry request after 1-2 second delay.

9.4 Client Troubleshooting Decision Tree



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10. Admin Control & Platform Administration Guide

PiyAPI provides a comprehensive **Admin Control Suite** for platform administrators, security operations teams, and customer support engineers.

10.1 Role-Based Access Control (RBAC) & Admin Authentication

Admin routes require an Admin JWT Session or Superuser API key. PiyAPI enforces four administrative role tiers:

Role Tier	Access Scope	Description
`founder`	Superuser (Cross-Tenant)	Full platform superuser. Can manage all tenants, SSO configs, and global settings.
`admin`	Tenant Admin	Admin access to user management, billing, quotas, and cache clearing.
`operator`	Operations / Support	Operator surface access (support impersonation, metrics monitoring).
`auditor`	Read-Only Compliance	Read-only access to audit logs, compliance reports, and legal holds.

```
Authorization: Bearer YOUR_ADMIN_JWT_TOKEN
```

10.2 User & Tenant Management (`/api/v1/admin/users`)

`GET /api/v1/admin/users` — List All Platform Users

- **Request:** `GET https://api.piyapi.cloud/api/v1/admin/users?page=1&limit=50&plan=pro&status=active`
- **Response Schema (200 OK):**

```
{
  "users": [
    {
      "id": "7c817a60-f566-4cd0-9046-3c8222f43126",
      "email": "developer@example.com",
      "name": "Alex Developer",
      "subscription_tier": "pro",
      "status": "active",
      "created_at": "2026-08-01T10:00:00.000Z"
    }
  ],
  "total": 1,
  "page": 1,
  "limit": 50
}
```

`GET /api/v1/admin/users/:id/usage` — Inspect Tenant Usage Metrics

- **Request:** `GET https://api.piyapi.cloud/api/v1/admin/users/7c817a60-f566-4cd0-9046-3c8222f43126/usage`
- **Response Schema (200 OK):**

```
{
  "user_id": "7c817a60-f566-4cd0-9046-3c8222f43126",
  "total_memories_stored": 4520,
  "total_search_queries": 12840,
  "total_tokens_consumed": 1850000,
  "active_connectors": 3,
  "byok_enabled": true
}
```

10.3 Support User Impersonation (`POST /api/v1/admin/users/:id/impersonate`)

Platform admins can generate a short-lived, forensically audited impersonation session token to reproduce client-reported issues directly in the context of that tenant.

- **Request Payload:**

```
{
  "reason": "Debugging search zero results bug reported in ticket #SUP-8812"
}
```

- **Response Schema (200 OK):**

```
{
  "impersonationToken": "eyJhbGciOiJIUzI1NiIsIn...",
  "expires_in_seconds": 3600,
  "target_user_id": "7c817a60-f566-4cd0-9046-3c8222f43126",
  "audit_log_id": "audit_991827..."
}
```

10.4 Plan & Quota Overrides (`PATCH /api/v1/admin/users/:id/plan`, `/api/v1/admin/enterprise-quotas`)

`PATCH /api/v1/admin/users/:id/plan` — Update User Subscription Plan

- **Request Payload:**

```
{
  "plan": "enterprise",
  "reason": "Enterprise custom contract upgrade approved"
}
```

- **Response Schema (200 OK):**

```
{
  "user_id": "7c817a60-f566-4cd0-9046-3c8222f43126",
  "new_subscription_tier": "enterprise",
  "updated_at": "2026-08-03T15:20:00.000Z"
}
```

10.5 Business Metrics & Cost Monitoring (`/api/v1/admin/metrics`)

`GET /api/v1/admin/metrics/business` — Platform Business Telemetry

- **Response Schema (200 OK):**

```
{
  "mrr_usd": 145000,
  "arr_usd": 1740000,
  "total_active_tenants": 1280,
  "plan_distribution": {
    "free": 850,
    "pro": 380,
    "enterprise": 50
  }
}
```

`GET /api/v1/admin/metrics/costs` — Provider LLM Cost Tracking

- **Response Schema (200 OK):**

```
{
  "total_llm_cost_mtd_usd": 12450.8,
  "by_provider": {
    "openai": 8200.5,
    "anthropic": 3150.3,
    "gemini": 1100.0
  }
}
```

10.6 Cache Management & Semantic Cache Purging (`/api/v1/admin/cache`)

- `POST /api/v1/admin/cache/clear`: Flushes all Redis query result caches.
- `POST /api/v1/admin/cache/clear/embedding`: Flushes semantic embedding vector cache.

10.7 Global AI Model Routing & PHI Rules Engine (`/api/v1/admin/ai`, `/api/v1/admin/phi-rules`)

- `POST /api/v1/admin/ai/config`: Set default system LLM and embedding model.
- `POST /api/v1/admin/phi-rules`: Register custom PHI regex detection patterns.

Client Developer & Enterprise User Guide — PiyAPI Cognitive Memory Platform.
